TravelFlow

Functional & Non-Functional Requirements

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1. Introduction & Document Purpose

This document defines the functional and non-functional requirements for TravelFlow — a digital platform for travel management, bus ticket booking, vehicle rental, and road transport fleet administration in the Republic of Albania. The requirements listed here are grounded in Albanian national legislation and have been developed in accordance with international software engineering standards.

TravelFlow serves three primary user categories:

- Clients (travellers) — passengers who search and book tickets
- Drivers (bus operators) — who manage assigned trips and validate tickets
- System Administrators — who oversee platform operations and reporting

Every requirement in this document is linked to a relevant legal basis and accompanied by its technical and legal implementation rationale.

Key Coverage Areas

- User management and access control
- Ticket booking and seat selection
- Vehicle rental
- Driver portal and ticket validation
- Fleet administration and maintenance
- Data protection and cybersecurity

Primary Legal References

Law	Subject
Law No. 124/2024	Personal Data Protection (GDPR-aligned)
Law No. 25/2024	Cybersecurity
Law No. 9918/2008	Electronic Communications in the Republic of Albania
Law No. 8308, dated 18.03.1998 (amended)	Road Transport

Law No. 55/2020	Payment Services

2. Functional Requirements

2.1 User Management & Authentication

FR-01 — New User Registration	
Legal Basis	*Law No. 9918/2008 — Electronic Communications*
Rationale	This law promotes the use of electronic systems for public services. An online registration system provides fast and easy access to the platform, reduces administrative burden, and complies with the legal framework supporting digital solutions.

FR-02 — User Authentication (Login)	
Legal Basis	*Law No. 124/2024 — Personal Data Protection*
Rationale	This law mandates strict measures for the protection of personal data. A secure authentication system protects user data from unauthorized access, ensuring the confidentiality and integrity of personal information.

FR-03 — Logout & Session Management	
Legal Basis	*Law No. 124/2024 — Personal Data Protection; Law No. 25/2024 — Cybersecurity*
Rationale	Data protection and cybersecurity laws require secure session management mechanisms. Logout functionality and automatic session expiry protect sensitive user data from unauthorized use.

FR-04 — Role-Based Access Control (RBAC)	
Legal Basis	*Law No. 124/2024 — Personal Data Protection*
Rationale	Implementing RBAC ensures that only authorized personnel have access to specific data, minimizing the risk of data breaches and ensuring compliance with data protection regulations.

2.2 Trip Search & Booking

FR-05 — Search Available Trips	
Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	Road transport law requires transparency in public transport schedules and pricing. A digital search engine provides instant access to all available trips, fulfilling transparency obligations and reducing administrative overhead.

FR-06 — Automatic Trip Generation (autoTrips)	
Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	To ensure continuity of transport service, the system automatically generates trips for any requested date. This fulfills legal requirements for continuous transport service provision and reduces manual scheduling errors.

FR-07 — Trip Card Display with Full Details	
Legal Basis	*Law No. 9918/2008 — Electronic Communications*
Rationale	The law promotes clear and complete electronic services. Displaying all trip details (route, timetable, bus model, price) ensures information transparency and helps clients

	make informed decisions.

2.3 Seat Selection & Booking Confirmation

FR-08 — Interactive Seat Selection	
Legal Basis	*Law No. 9918/2008 — Electronic Communications*
Rationale	Providing an interactive seat map ensures service transparency, gives the client freedom of choice, and complies with e-commerce principles for providing clear information before confirming a transaction.

FR-09 — Dynamic Price Calculation by Destination	
Legal Basis	*Law No. 55/2020 — Payment Services*
Rationale	Payment services law requires full price and fee transparency prior to confirmation. Dynamic price calculation based on the chosen destination ensures full legal compliance and prevents cost misinterpretation.

FR-10 — Full Fare Breakdown Before Confirmation	
Legal Basis	*Law No. 55/2020 — Payment Services*
Rationale	Under payment services law, the consumer must be clearly informed of all costs before completing a transaction. Displaying ticket price, service fee, and seat fee itemised guarantees full financial transparency.

FR-11 — Booking Confirmation with Unique Code

Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	Issuing a unique confirmation document for each booking fulfills legal requirements for travel documentation and ensures proof of transaction for both parties.

FR-12 — Random Seat Booking	
Legal Basis	*Law No. 9918/2008 — Electronic Communications*
Rationale	The option of a random seat without additional fees increases service accessibility and complies with e-service principles of being accessible to all users, without discrimination based on ability to pay extra charges.

2.4 Booking History & Digital Tickets

FR-13 — View Booking History	
Legal Basis	*Law No. 124/2024 — Personal Data Protection*
Rationale	The law guarantees the user's right to access their own personal data. A booking history section ensures full transparency and compliance with data subject rights.

FR-14 — Digital Ticket with Full Details	
Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	A digital ticket is a legal travel document. It must contain all data necessary to identify the traveller, route, date, and seat assignment.

FR-15 — Booking Cancellation

Legal Basis	*Law No. 55/2020 — Payment Services; Law No. 9918/2008 — Electronic Communications*
Rationale	The right to return and cancel unused services is provided for in payment services legislation. A transparent cancellation function with confirmation dialog protects both parties.

2.5 Vehicle Rental

FR-16 — Catalogue of Available Rental Vehicles	
Legal Basis	*Law No. 9918/2008 — Electronic Communications*
Rationale	Displaying a complete vehicle catalogue with search filters complies with requirements for transparency in commercial electronic services. Clients must have full access to all information before completing a booking.

FR-17 — Vehicle Booking with Total Price Confirmation	
Legal Basis	*Law No. 55/2020 — Payment Services*
Rationale	Confirming the total rental price before finalising the transaction is a legal obligation under payment services legislation. The system automatically calculates the number of days and total price.

FR-18 — Rental History & Status	
Legal Basis	*Law No. 124/2024 — Personal Data Protection*
Rationale	Data subject rights include the right of access to personal transaction history. The

	rental history section ensures compliance with these rights.

FR-19 — Vehicle Return & Availability Reset	
Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	Documentation of vehicle return is mandatory for vehicle rental operators. The system automatically updates the vehicle status upon return.

2.6 Driver Portal

FR-20 — Driver Portal Access	
Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	Bus drivers need access to their assigned trip information. A dedicated portal ensures drivers only have access to information relevant to their role, fulfilling the operational requirements of road transport.

FR-21 — Access Isolation — Driver Records Only	
Legal Basis	*Law No. 124/2024 — Personal Data Protection*
Rationale	The data minimisation principle requires that each user has access only to data necessary for their role. Drivers must not have access to administrative or financial data.

2.7 Ticket Validation

FR-22 — Ticket Code Validation	
Legal Basis	*Law No. 8308 (amended) — Road Transport; Law No. 9918/2008*
Rationale	Ticket checking is a legal obligation for public transport operators. A digital validation system speeds up the boarding process, eliminates counterfeit tickets, and ensures precise passenger records.

FR-23 — Display Passenger Information After Validation	
Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	Correct identification of the passenger and their seat is necessary for safe bus operations. Displaying name, seat, station, and route helps drivers direct passengers efficiently.

FR-24 — Detection of Invalid & Duplicate Tickets	
Legal Basis	*Law No. 25/2024 — Cybersecurity; Law No. 8308*

FR-25 — Continuous Validation — No Page Refresh

Legal Basis	*Law No. 9918/2008 — Electronic Communications*
Rationale	The efficiency of the boarding process is important to maintaining road transport schedules. A fast, continuous interface ensures the boarding process is completed within the time defined by official timetables.

2.8 Admin Panel

FR-26 — Admin Panel Access	
Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	Transport operators must have a consolidated view of all operations. The admin panel enables effective platform oversight and fulfills reporting obligations to competent authorities.

FR-27 — Bus Fleet Management	
Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	Registration and management of the vehicle fleet is a legal obligation for road transport operators. The system maintains complete digital records for every vehicle.

FR-28 — Rental Fleet Management	
Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	Registration of rental vehicles and tracking of their availability complies with

	requirements for transparency in the provision of rental services.

FR-29 — User Management — View All	
Legal Basis	*Law No. 124/2024 — Personal Data Protection*

FR-30 — System Statistics — Read-Only View	
Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	Transport operators are required to maintain statistical records of their activity. A read-only statistics panel ensures data integrity by preventing accidental modifications.

2.9 Vehicle Maintenance Management

FR-31 — View Maintenance Schedule	
Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	Road transport law requires that vehicles be in good technical condition and undergo periodic inspections. A digital maintenance system ensures continuous monitoring and compliance with these obligations.

FR-32 — Mark Services as Completed	
Legal Basis	*Law No. 8308 (amended) — Road Transport*
Rationale	Digital documentation of completed services provides precise records for official inspections and automatically calculates future service dates, reducing the risk of neglected maintenance.

3. Non-Functional Requirements

3.1 Security Requirements

Requirement	**Encryption of sensitive user data at rest and in transit.**
Legal Basis	*Law No. 124/2024 — Personal Data Protection (GDPR-aligned)*
Rationale	This law mandates personal data protection in line with GDPR standards. Data encryption ensures that sensitive information remains confidential and protected from unauthorized access.

Requirement	**Multi-Factor Authentication (MFA) for administrative users.**
Legal Basis	*Law No. 25/2024 — Cybersecurity*
Rationale	The cybersecurity law emphasizes the need for strong authentication mechanisms to prevent unauthorized access. MFA implementation increases security by requiring multiple verification methods.

Requirement	**Role-Based Access Control (RBAC) to restrict unauthorized access.**
Legal Basis	*Law No. 124/2024 — Personal Data Protection*
Rationale	Implementing RBAC ensures that only authorized personnel access specific data, minimizing breach risks and ensuring compliance with data protection regulations.

Requirement	**Periodic penetration testing and security audits.**
Legal Basis	*Law No. 25/2024 — Cybersecurity*
Rationale	Regular security assessments help identify and mitigate vulnerabilities, ensuring system resilience against cyber threats and compliance with legal requirements for maintaining cybersecurity standards.

Requirement	**Secure backup mechanisms to prevent data loss.**
Legal Basis	*Law No. 25/2024 — Cybersecurity*
Rationale	Secure backup solutions ensure data integrity and availability, enabling rapid recovery in the event of cyber incidents, in line with legal mandates for data protection.

Requirement	**GDPR compliance and compliance with local data protection laws.**
Legal Basis	*Law No. 124/2024 — Personal Data Protection (GDPR-aligned)*
Rationale	Respecting GDPR and local data protection laws ensures that personal data is processed lawfully, transparently, and securely, protecting individual rights and maintaining institutional integrity.

3.2 Performance Requirements

Requirement	**System response time under 3 seconds for standard operations.**
Legal Basis	*Law No. 9918/2008 — Electronic Communications*

Rationale	Ensuring fast system response enhances the user experience and aligns with the law's aim of promoting efficient electronic communications.

Requirement	**High availability — at least 99.9% uptime during peak activity periods.**
Legal Basis	*Law No. 9918/2008 — Electronic Communications*
Rationale	Maintaining high system availability ensures continuous access to transport resources, supporting the law's goal of promoting reliable communication services.

Requirement	**Support for at least 500 simultaneous users without performance degradation.**
Legal Basis	*Law No. 9918/2008 — Electronic Communications*
Rationale	Accommodating a large number of simultaneous users ensures system scalability and aligns with legal principles promoting the development of robust electronic communication networks.

3.3 Reliability Requirements

Requirement	**Automated backups to ensure minimal data loss in the event of failures.**
Legal Basis	*Law No. 25/2024 — Cybersecurity*

Requirement	**Redundancy and failover mechanisms for database availability.**
Legal Basis	*Law No. 25/2024 — Cybersecurity*
Rationale	Ensuring system redundancy minimizes downtime and protects against data loss, meeting the cybersecurity standards prescribed by law.

3.4 Accessibility & Usability

Requirement	**The user interface must be intuitive and usable without prior training.**
Legal Basis	*Law No. 9918/2008 — Electronic Communications*
Rationale	A simple and intuitive user interface ensures that the service is accessible to every citizen regardless of their digital literacy level, promoting digital inclusion.

Requirement	**Support for multiple platforms — desktop browsers and mobile devices.**
Legal Basis	*Law No. 9918/2008 — Electronic Communications*
Rationale	Providing access to the service across different platforms ensures service availability for all citizens, fulfilling the principles of universal access to electronic services.

Requirement	**Clear and understandable error messages for all failure scenarios.**
Legal Basis	*Law No. 9918/2008 — Electronic Communications*
Rationale	Clear error communication protects users from erroneous transactions and ensures that the electronic service is transparent and trustworthy according to legal standards.

4. Transitional Provisions

The requirements set out in this document come into effect immediately upon official approval by the competent authorities. Any subsequent changes to Albanian legislation or applicable international standards will be reflected in a new version of this document.

The TravelFlow development team commits to periodically reviewing the system's compliance with current legislation and implementing necessary changes within the deadlines set by the competent authorities.

Document Author	Technical Director	Legal Approver
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Date: _____ / _____ / _____	Date: _____ / _____ / _____	Date: _____ / _____ / _____